Automating Parametric CAD Sketch Synthesis via Sequential Decision Making

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1. Goal

Modern mechanical design teams still spend significant time translating requirement tables into parametric computer-aided design (CAD) sketches that seed downstream solid modeling. I propose to automate the early-stage sketching of planar bracket components directly from structured requirement tuples by learning from the SketchGraphs subset of the Fusion 360 Gallery dataset, which provides 15,390 human-authored sketches with fully timestamped modeling operations, constraint graphs, and parameter values (Autodesk Research, 2021, 2020). Focusing on L- and U-shaped brackets with at most 12 primitives focuses on exercising sequential decision-making tools on meaningful engineering constraints.

We have three deliverables: (i) Days 1–4: curate a bracket-focused dataset slice, extract feature statistics (primitive frequencies, constraint co-occurrence, dimensional ranges), and publish preprocessing notebooks; (ii) Days 5–9: train and validate a behavior-cloned policy that maps requirement tokens and partial sketches to the next CAD action, logging calibration metrics for downstream risk analysis; and (iii) Days 10–14: layer an uncertainty-aware action selection module (conservative Q-learning fine-tuning plus ensemble-based confidence scoring) and run ablation studies quantifying constraint satisfaction rate, sketch length, and solver success on a held-out specification set. Achieving these milestones will demonstrate that principled decision-making approaches can materially reduce iteration time in CAD automation while respecting design-engineering rigor (Kochenderfer et al., 2022).

2. Decision Making

I frame automated sketch construction as a partially observable Markov decision process (POMDP) in which each episode corresponds to building a parametric sketch that satisfies a requirement tuple (g, c, d) containing desired geometry (hole pattern, flange lengths), hard constraints (mounting plane, thickness), and design intents (load directions). The latent state s_t comprises the full canonical sketch graph, including primitive embeddings, unresolved constraint slack, and the latent "style cluster" inferred from SketchGraphs metadata (Autodesk provides eight unsupervised clusters covering structural, decorative, and mechanical sketches (Autodesk Research, 2020)). The observation o_t exposes the partial sketch graph available to the agent, requirement tokens, and noisy signals from the constraint solver (feasibility flags, residual norms).

The action space leverages SketchGraphs' taxonomy: five primitive types (line, arc, circle, point, spline) and nine constraint/dimension types (coincident, perpendicular, parallel, equal, tangent, horizontal, vertical, distance, angle) with parameter bins learned from empirical quantiles. Each action is factored into a discrete intent a_t^{type} and continuous parameter vector a_t^{param}; the former is modeled as a categorical distribution, whereas the latter is sampled from Gaussian heads centered on dataset means with learned variance.

Transitions $T(s_{t+1} \mid s_t, a_t)$ are deterministic under the CAD kernel but I model tolerance-related perturbations via a learned residual dynamics model that perturbs endpoints and constraint slack to mimic solver adjustments.

Rewards combine dense shape guidance with terminal feasibility checks. A -0.05 step cost encourages concise sketches, partial rewards score constraint satisfaction and similarity between generated and target dimensions (scaled Chamfer distance on anchor points), and a terminal bonus of $+1.0$ is granted when all hard constraints fall within tolerance bands. The hierarchical structure enables efficient planning: a high-level policy selects which substructure to extend (e.g., "extend flange") while a low-level controller chooses primitive parameters. Offline behavior cloning on filtered expert trajectories supplies the initial policy; uncertainty-aware improvement follows via conservative Q-learning and limited-horizon model-predictive control that rolls out candidate action sequences under the learned residual model.

3. Sources of Uncertainty

Three uncertainty sources dominate this problem. (i) **Demonstration variability:** SketchGraphs encodes multiple valid action permutations for similar brackets; some designers add driving dimensions before closing loops, others after. I will capture this epistemic uncertainty with a bootstrap ensemble of graph encoders and Q-heads, using disagreement as a risk metric. (ii) **Specification ambiguity:** Requirement tuples derived from handbooks may omit secondary dimensions (fillet radii, hole clearances). I will model the missing values as latent random variables with truncated Gaussian priors fit to the curated dataset slice and maintain beliefs β_t that are updated whenever the solver reports infeasible constraints. (iii) **Solver stochasticity and tolerances:** Autodesk's constraint solver occasionally perturbs primitive endpoints when loops are nearly singular, effectively injecting action noise. I will emulate this aleatoric component through the learned residual dynamics model and propagate tolerance bands (e.g., ± 0.2 mm for bolt-hole spacing) into the reward.

During training, Monte Carlo dropout and deep ensembles yield calibrated predictive intervals; conservative Q-learning uses the lower confidence bound to penalize risky actions. At deployment, the agent selects actions using a risk-sensitive criterion $\max_a \mathbb{E}[Q(s,a)] - \lambda\sqrt{\mathrm{Var}[Q(s,a)]}$, where λ is tuned on validation requirements to balance aggressiveness and robustness. Evaluation will report mean performance, 95% confidence intervals, and worst-case outcomes conditioned on solver failures, providing a principled view of residual uncertainty.

4. Sketches of Solution (Optional)

Figure 1 depicts the closed-loop decision process the agent will implement, highlighting how specification parsing, belief updates, policy inference, and uncertainty handling interact with the CAD solver.

References

Autodesk Research. Sketchgraphs: A large-scale dataset for modeling relational geometry

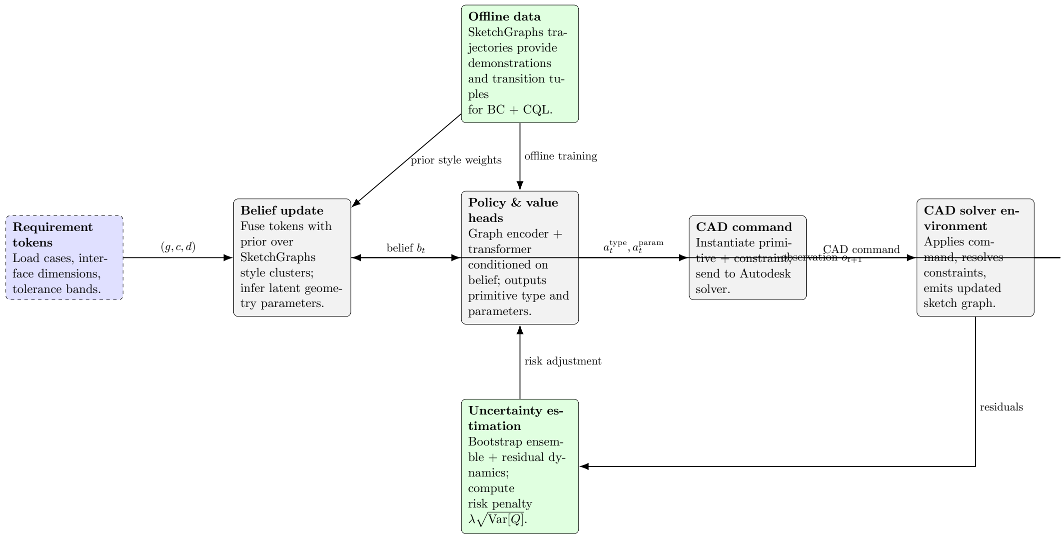

Figure 1: Algorithmic loop: requirements feed a belief-augmented policy trained on SketchGraphs; uncertainty estimates shape action selection before commands are executed in the CAD solver, whose feedback updates both belief and risk models.

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